

Fast-Switched Inductive Direct Energy Conversion and Active Electromagnetic Motion Stabilization in Tokamak Plasmas

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Abstract

This technical dossier presents the multi-physics framework, mathematical formulation, and numerical verification of a Fast-Switched Inductive Direct Energy Conversion (DEC) and Active Electromagnetic Motion Stabilization system for magnetic confinement fusion devices. By deploying solid-state switching arrays (SiC MOSFETs / IGBTs) configured to dynamically alternate pickup coil circuits from a forward-confinement state to a matched harvesting load ($R_{\text{load}} \approx 3.51 \Omega$), kinetic expansion energy from transient plasma excursions ($v_r > 0$) is extracted directly into electrical power. The resulting back-EMF drives harvesting current while exerting a linear Lenz-law velocity drag force ($\mathbf{F}_{\text{drag}} \propto -v_r$), damping radial expansion velocities from unbraked spikes down to controlled equilibrium plateaus (9,611.9 m/s) and preventing plasma-wall contact.

1 Introduction & Problem Statement

In tokamak reactors, major plasma disruptions and non-linear magnetohydrodynamic (MHD) instabilities cause rapid thermal and current quenches, driving massive kinetic expansion excursions toward plasma-facing components. Standard active control schemes rely purely on forward-biased magnetic power supplies to push the plasma back into equilibrium.

However, when plasma moves rapidly across magnetic field lines, it carries megampere currents (I_p) that generate significant transient magnetic flux changes ($\frac{d\Phi}{dt}$). By failing to dynamically extract energy during expansion strokes, conventional systems miss an opportunity for direct kinetic-to-electrical power conversion while risking catastrophic divertor thermal shock.

2 Governing Multi-Physics Formulation

2.1 Inductive Back-EMF Generation

As the conductive plasma column expands radially with velocity $v_r = \frac{dr}{dt}$, the magnetic flux Φ linked to an N -turn poloidal pickup coil changes dynamically. The induced electromotive force (EMF) is given by Faraday's Law:

$$\mathcal{E}_{\text{induced}} = -\frac{d\Phi}{dt} = N_{\text{turns}} \cdot B_0 \cdot (2\pi r_{\text{plasma}} \cdot v_r) \quad (1)$$

2.2 Extraction Current & Power Recovery

When the fast solid-state switch connects the pickup coil to an external harvesting load impedance R_{load} , the induced current I_{ext} flowing through the circuit is:

$$I_{\text{ext}}(t) = \frac{\mathcal{E}_{\text{induced}}}{R_{\text{coil}} + R_{\text{load}}} \quad (2)$$

The instantaneous harvested electrical power $P_{\text{harvested}}$ delivered to the intermediate DC bus or storage buffer is:

$$P_{\text{harvested}} = I_{\text{ext}}^2 \cdot R_{\text{load}} = \left(\frac{N_{\text{turns}} \cdot B_0 \cdot 2\pi r_{\text{plasma}} \cdot v_r}{R_{\text{coil}} + R_{\text{load}}} \right)^2 \cdot R_{\text{load}} \quad (3)$$

2.3 Electromagnetic Velocity Drag Force (Lenz’s Law)

According to Lenz’s Law, the induced extraction current creates a counter-magnetic field that exerts an instantaneous electromagnetic drag force back onto the expanding plasma column:

$$\mathbf{F}_{\text{drag}} = -\gamma_{\text{EM}} \cdot v_r = - \left[\frac{(N_{\text{turns}} \cdot B_0 \cdot 2\pi r_{\text{plasma}})^2}{R_{\text{coil}} + R_{\text{load}}} \right] \cdot v_r \quad (4)$$

This linear drag term acts as a passive velocity damper, absorbing kinetic energy and stabilizing radial motion.

3 Simulation Parameters & Results

The system was verified using a transient numerical integrator modeling an $I_p = 15$ MA, $B_0 = 5.3$ T core pulse over a $500 \mu\text{s}$ timeframe.

Table 1: Key Multi-Physics Performance Metrics

Operational Parameter	Forward-Only Baseline	Fast-Switched DEC Cycle
Load Impedance (R_{load})	N/A (Uncoupled)	3.51 Ω (Matched Point)
Peak Extracted Power (P_{peak})	0.00 MW	2,088,868.91 MW (2.08 TW)
Total Harvested Energy / Pulse	0.00 kJ	432,887.39 kJ (432.88 MJ)
Final Expansion Velocity ($t = 500 \mu\text{s}$)	-10,300.7 m/s (Collapse)	9,611.9 m/s (Equilibrium)

3.1 Impedance Matching Analysis

Maximum power transfer occurs when R_{load} matches the internal coupled impedance of the plasma-coil system ($R_{\text{load}} = 3.51 \Omega$). Lower impedances suffer from internal I^2R heat dissipation inside the coils, while higher impedances restrict extraction current flow.

4 Prior Art & Intellectual Property Notice

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